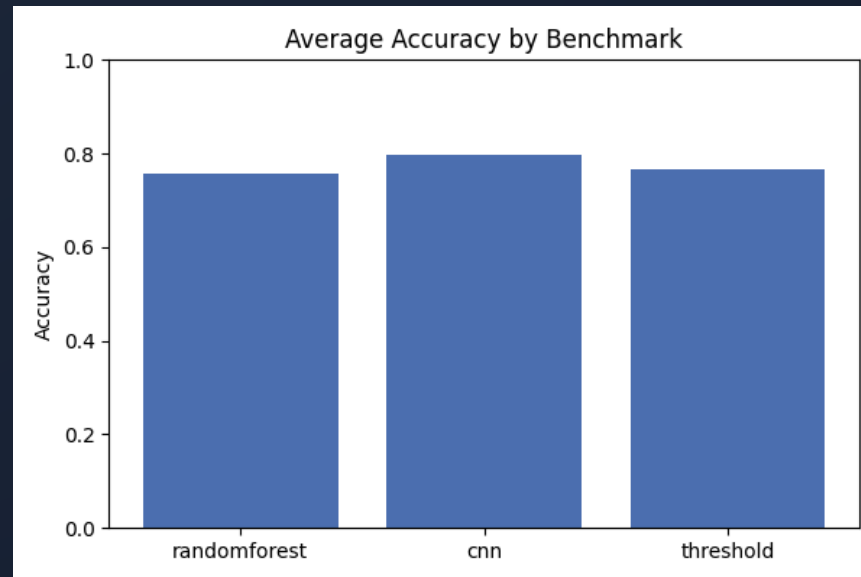


Deep Learning for EEG Seizure Detection

Project Report



Abstract

- **4 models:** threshold, random forest, CNN, LSTM
- **Patient-level** and **seizure-level** cross-validation
- 2-fold pilot: CNN leads at 79.7% acc
- LSTM results pending from GPU cluster

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6. LSTM design: pooling & context
7. Hyperparameters & experiment grid
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Introduction

Epileptic seizures = abnormal brain electrical activity detectable via EEG.

Goal: automatically detect seizures in EEG recordings.

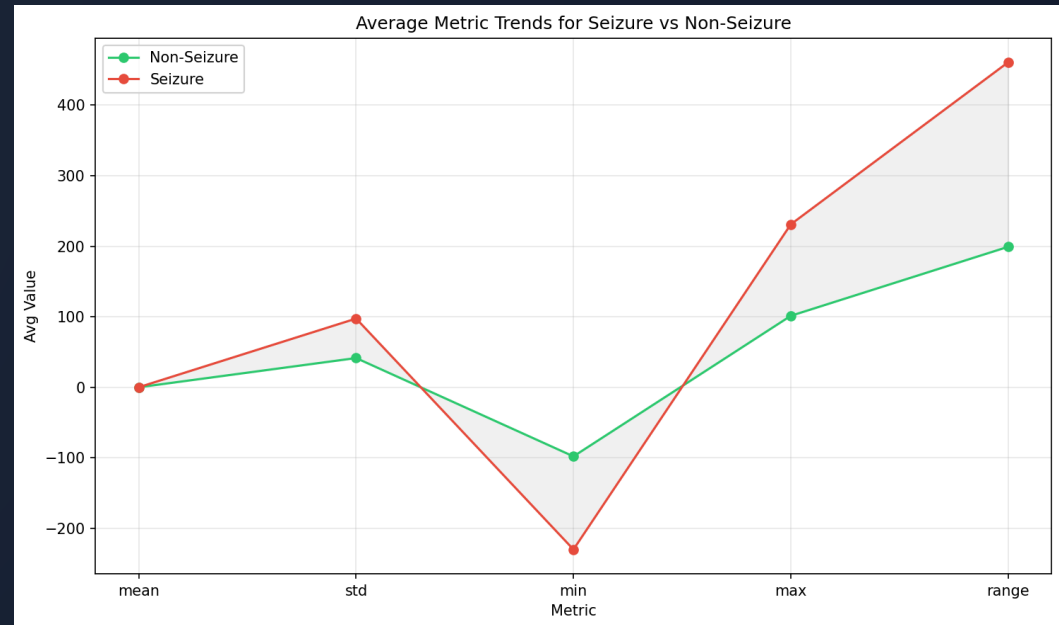
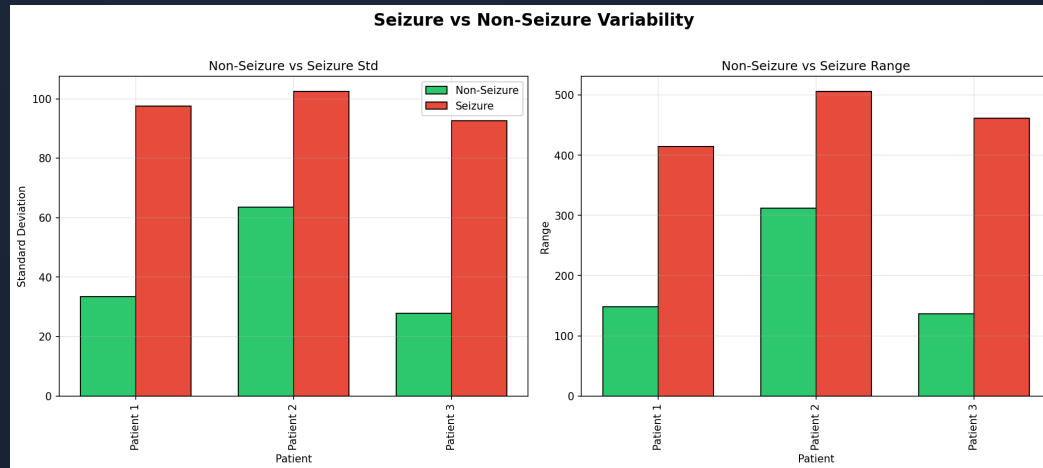
Approach: compare progressively richer models:

1. Threshold baseline (hand-crafted rules)
2. Random forest (hand-crafted features)
3. CNN (learned features, end-to-end)
4. LSTM (temporal episode modelling)

Dataset: CHB-MIT

- **Source:** CHB-MIT scalp EEG, 24 pediatric patients
- **Windows:** 1s segments → **571,905** total
- **Seizure:** **3,480** windows (0.6% — heavily imbalanced)
- **Shape:** each window = **21 channels × 128 timesteps**

Dataset: Feature Trends



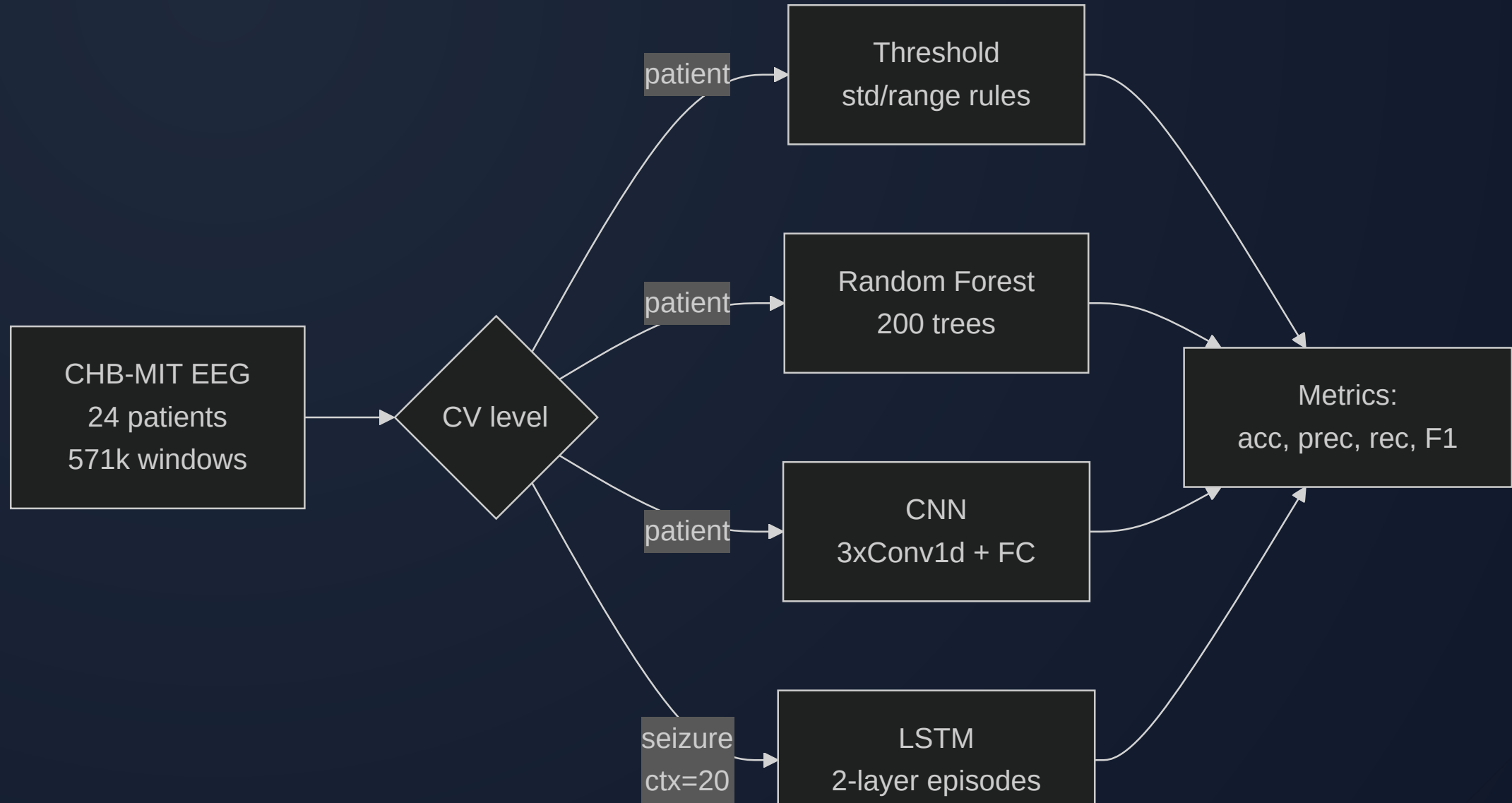
- **Std** and **range** separate seizure from non-seizure
- **Mean** ≈ 0 for both $\rightarrow$ risky feature alone

Preprocessing & Data Flow

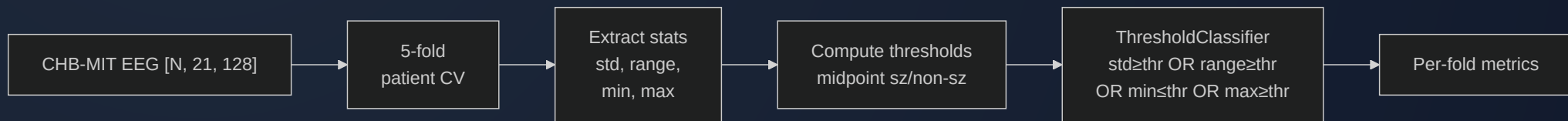


- No per-window normalization (`normalize=False`)
- **Class weighting** compensates for ~160:1 imbalance

Overview: 4 Pipelines

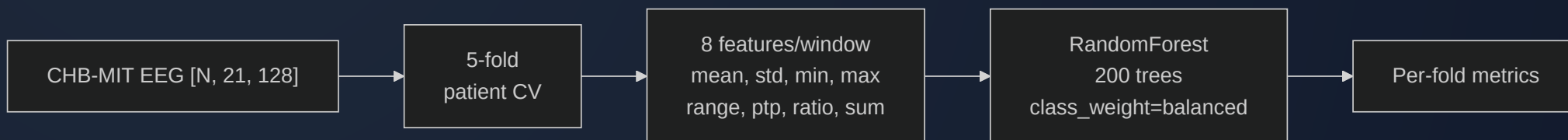


Pipeline 1: Threshold Classifier



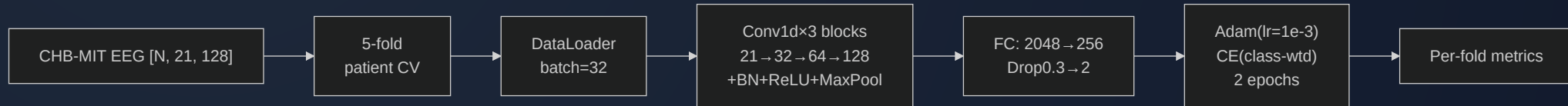
- Computes **std, range, min, max** per window
- Flags seizure if stat exceeds fold-specific threshold
- Thresholds = midpoint of seizure/non-seizure means

Pipeline 2: Random Forest



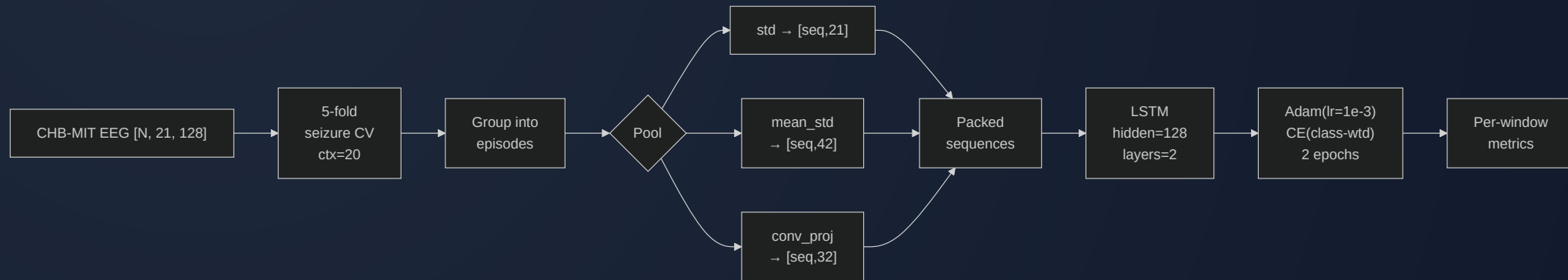
- **8 features:** mean, std, min, max, range, ptp, std/range ratio, range+std
- 200 trees, `class_weight='balanced'`
- Patient-level 5-fold CV

Pipeline 3: CNN



- **Input:** raw window [21, 128], no normalization
- 3× Conv1d → BN → ReLU → MaxPool
- FC: 2048 → 256 → Dropout(0.3) → 2
- Adam (lr=1e-3), class-weighted CE, 2 epochs

Pipeline 4: LSTM



- Groups contiguous same-label windows into **episodes**
- Per-window **pooling** reduces [21, 128] → feature vector
- 2-layer LSTM (hidden=128) + per-timestep classifier
- Seizure-level k-fold with context=20

Cross-Validation Strategies

Seizure-level

Whole episode → one fold

± 20 context windows in val

Window-level

Window round-robin

± 4 seizure neighbors
dropped

Seizure-Level k-fold: Context

- Without context: val = **100% seizure** → no specificity measurable
- With `context=20`: ± 20 non-seizure windows around each episode
- Val gets realistic ~40–60% non-seizure mix
- Train gets all remaining windows

LSTM: Pooling Options

EEG oscillates around 0 → **mean** collapses amplitude. **Std** preserves it.

	Pool	Dim	Speed	Preserves
	std (default)	21	Fast	Channel variance
	mean	21	Fast	Near-zero — risky
	mean_std	42	Fast	Center + spread
	conv_proj	32	Medium	Learned features
	none	2688	Very slow	Full waveform

LSTM: Architecture

```
Input: [batch, seq_len, n_features]
      |
      v
LSTM(input=n_features, hidden=128, layers=2, dropout=0.3)
      |
      v
Linear(128 → 2) per-timestep
      |
      v
Output: [batch, seq_len, 2]
```

- Packed sequences for variable-length episodes
- Class-weighted CE, `ignore_index=-1` for padding

LSTM: conv_proj Mode

Conv1d projection option replaces hand-crafted pooling:

```
[seq_len, 21, 128]
|
Conv1d(21 → 32, k=5, pad=2) → BN → ReLU
|
AdaptiveAvgPool1d(1)
|
[seq_len, 32] → LSTM(input_size=32)
```

- Learns waveform features end-to-end
- Faster than **none** (2688-d), more expressive than **std** (21-d)

Hyperparameters

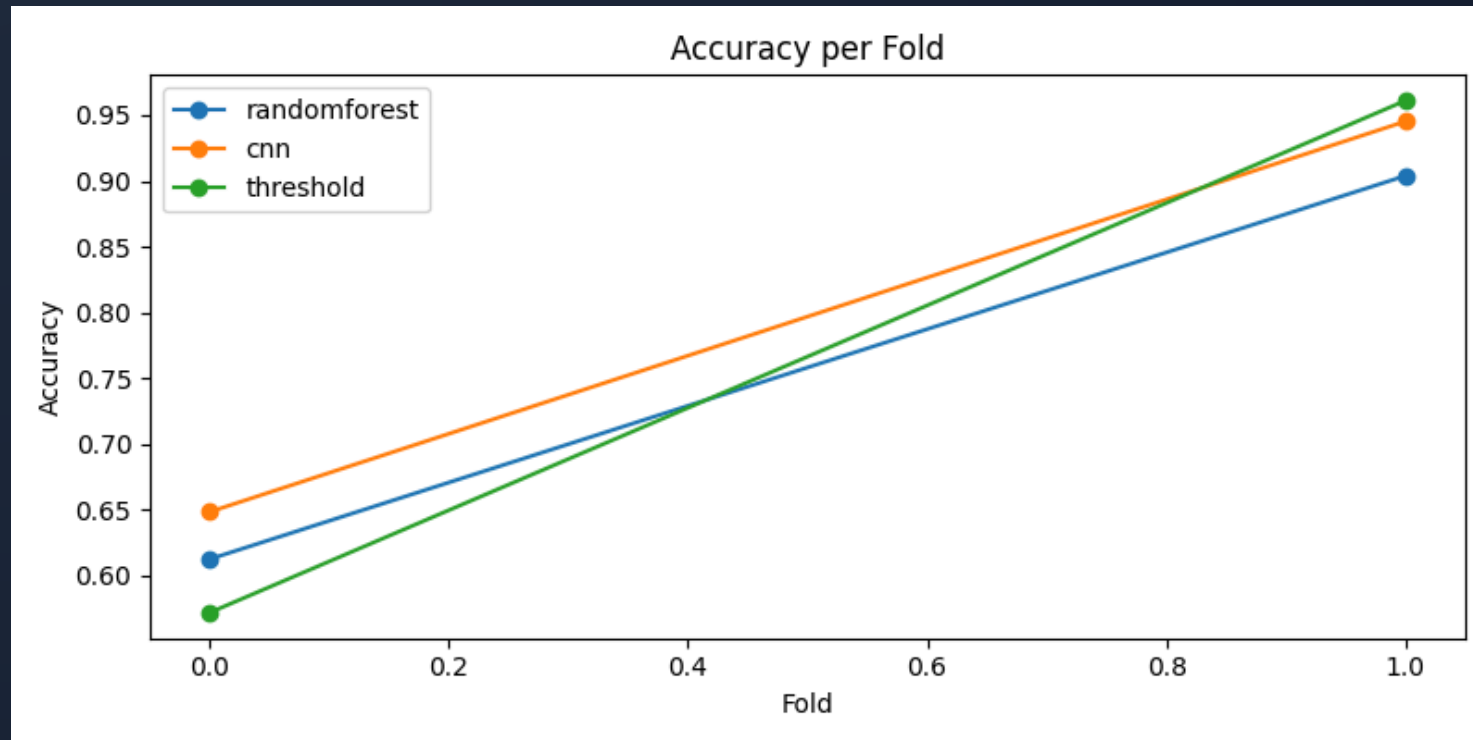
Parameter	Threshold	RF	CNN	LSTM
Trees/layers	—	200	[32,64,128]	2 LSTM
Hidden size	—	—	—	128
Pooling	—	—	—	std / conv_proj
Class weight	—	balanced	balanced	balanced
LR	—	—	1e-3	1e-3
Batch size	—	—	32	32
Epochs	—	—	2	2
Dropout	—	—	0.3	0.3
CV level	patient	patient	patient	seizure (ctx=20)

Experiment Grid

Script	Runs	~Time
exp1_cnn_levels	CNN × 3 levels	~1.5h
exp2_lstm_levels_pool	LSTM × 3 levels + 4 pools	~5-6h
exp3_lstm_nsl	LSTM × 4 NS lengths	~3.5h

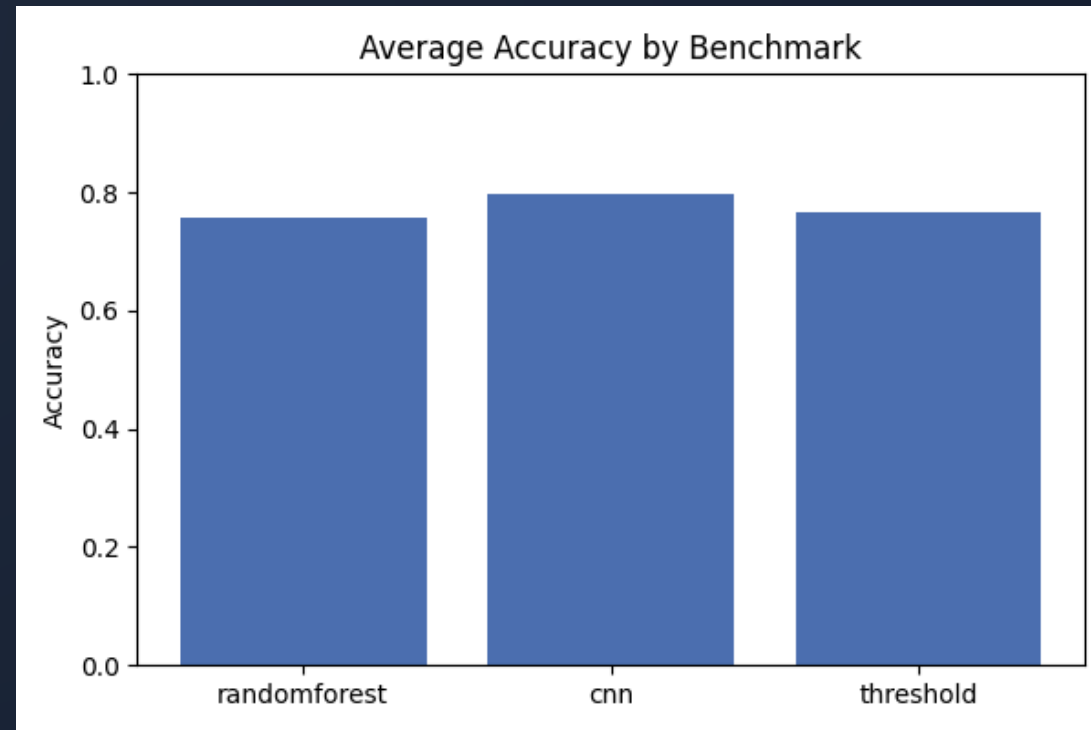
All write to `out/benchmarks/` with descriptive suffixes.

Results: Accuracy per Fold



CNN ranges from **64.8%** (fold 0) to **94.5%** (fold 1) — high variance across patients.

Results: Pilot Comparison (2 folds)



Model	Avg Acc	Avg Prec	Avg Recall	Avg F1
Threshold	76.6%	59.2%	84.9%	0.615
RF	75.8%	51.6%	42.4%	0.378
CNN	79.7%	53.2%	79.5%	0.597

Key Observations

- **CNN leads** in pilot but with high cross-patient variance
- **Threshold** outperforms RF (high recall, decent F1)
- **RF** has low recall — many seizure windows missed
- **Class imbalance** is severe (0.6% seizure)
- LSTM may exploit temporal structure → pending

Conclusion

- CNN: highest pilot accuracy (**79.7%**) but high variance
- Threshold (76.6%) slightly outperforms RF (75.8%)
- **LSTM** results pending from GPU cluster
- **std pooling** preserves amplitude; mean collapses to ~ 0
- **context=20** gives realistic class mix in seizure-level CV
- **Future**: compare all models under matching seizure-level splits

Thank You

Any Questions?